Tourism Risk & Context Intelligence System: AI-Based Tourism Risk Prediction & Advisory Platform for Sri Lanka Tourism

Project ID – R26-IT-152

Project Proposal Report

Abinaya R — IT22090058.

Bachelor of Science (Hons) Degree in Information Technology Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka (SLIIT)

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ABSTRACT

In 2025, Sri Lanka recorded 2,360,000 international arrivals, with a national target of three million by the end of 2026. Despite this growth, the country lacks an integrated, site-level risk prediction and context intelligence system for tourist destinations. This gap contributes directly to overcrowding at popular sites, tourist safety incidents, and suppressed per capita daily spending, which at approximately USD 148 remains well below regional benchmarks.

This research proposes a web-based Tourism Risk & Context Intelligence System that integrates four data sources — flight arrival data from the AviationStack API, regional hotel occupancy metrics, historical visitor statistics from the Sri Lanka Tourism Development Authority (SLTDA), and weather data from OpenWeatherMap — into a Random Forest machine learning pipeline. The system enables tourists to assess crowd density and contextual risk at tourist sites in advance, supporting safer trip planning and improved spending outcomes.

The system is designed to serve 500+ registered tourist destinations across Sri Lanka. Where hourly data is unavailable, intra-day crowd and risk patterns are simulated using proxy indicators and clearly labelled as such in all outputs. Through the platform, tourists receive risk-level classifications (Low / Medium / High), recommendations for optimal visiting times, near real-time overcrowding alerts, and the ability to dynamically adjust their itineraries in response to predicted risk and crowd conditions. When a destination is forecast to be heavily congested or high-risk, the system suggests environmentally-friendly alternative sites to redistribute visitor demand.

The expected outcomes include a predictive model achieving MAE ≤ 0.25 and $R^2 \geq 0.7$ on normalised crowd score prediction, an improved safety experience for tourists visiting Sri Lanka, and actionable intelligence for tourism operators and policymakers. The platform aligns with SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 11 (Sustainable Cities and Communities). Commercially, the platform is structured around three revenue streams: a freemium model for individual tourists, a B2B SaaS dashboard for tourism operators, and an anonymised data licensing model for institutions such as SLTDA.

Keywords: Risk Intelligence, Crowd Prediction, Random Forest, Context-Aware Advisory, Dynamic Itinerary Adjustment, Green Tourism, Sri Lanka

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Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material previously published or written by another person, apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

only for Proposal 14/03/2026

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Signature of the Supervisor

Mr. Buddika Harshanath

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Date

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Signature of the Co-Supervisor

Prof. Samantha Rajapaksa

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Date

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
API	Application Programming Interface
SLTDA	Sri Lanka Tourism Development Authority
CMB	Bandaranaike International Airport IATA Code
MAE	Mean Absolute Error
POI	Point of Interest
SDG	Sustainable Development Goal
WBS	Work Breakdown Structure
REST	Representational State Transfer
SaaS	Software as a Service
DSR	Design Science Research
SHAP	SHapley Additive exPlanations
NLP	Natural Language Processing

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INTRODUCTION

1.1 Background and Context

The tourism sector of Sri Lanka has made a significant contribution to the country's foreign exchange income, with 2.36 million international arrivals in 2025 and plans to reach 3 million by 2026 [1]. While this growth represents a major economic opportunity, it also reveals a critical operational gap: the absence of real-time, site-specific risk and crowd intelligence to manage tourism at destination level. Many of the most popular heritage and natural sites — such as Sigiriya Rock Fortress, Galle Fort, Yala National Park, Mirissa Beach, and Kandy's Temple of the Tooth — currently operate without any data-driven intelligence regarding expected visitor volumes, contextual risk factors, or crowd conditions on any given day.

Without such intelligence, tourists visiting these sites rely solely on static historical data from general-purpose platforms and are unable to assess how crowded or risky a destination will be before they arrive. As a result, visitors experience overcrowding, excessive wait times, reduced access to attractions, and lowered satisfaction — contributing to Sri Lanka's average daily tourist spending of approximately USD 148, which is well below the regional benchmark of USD 300 [1]. In addition, unmanaged visitor congestion at UNESCO-listed sites increases the rate of structural and environmental degradation, directly undermining the objectives of Sustainable Development Goal 11.4.

Tourism operators also suffer from the absence of predictive tools. Hotels, guiding companies, and experience providers cannot anticipate high-traffic periods, resulting in understaffing, inconsistent service quality, and missed revenue opportunities. According to the SLTDA 2024 Year-in-Review report, there is a shortfall of over 1,600 registered guides relative to current tourism demand [1].

Critically, existing platforms provide no mechanism to predict whether a tourist site will be safe, crowded, or contextually appropriate for a visit on a given day. They do not account for local events, public holidays, flight arrival surges, or weather-driven behavioural shifts that directly affect crowd levels and tourist safety. This absence of context-aware risk and crowd intelligence is the core problem this research addresses.

Analysis of Existing Systems and Their Limitations

Currently, the platforms available to Sri Lankan tourists lack predictive and contextual relevance, as Table 1 demonstrates by comparison of five current system limitations for Sri Lanka.

Table 1: Breakdown of Existing System Limitations for Sri Lankan Tourism

Platform	Core Capability	Critical Limitation for Sri Lanka
Google Maps Popular Times	Database of historical averages for each hour of the day	Only Backwards Looking - No predictive insight into future busyness; flight and hotel data not integrated; no Alerts or Itinerary management.

Platform	Core Capability	Critical Limitation for Sri Lanka
TripAdvisor	Community reviews and ratings of destinations Static System	Static system; no predictive insight; no risk scoring; no alerts or itinerary management capabilities
Sri Lanka Tourism App (SLTDA)	Site descriptions, opening hours, directions	No crowd intelligence, no risk assessment, no safety features
Booking.com / Expedia	Hotel & flight booking Primarily logistic solution	Primarily logistics; no crowd intelligence or site-level risk features
Google's Waze (crowd routing)	Real-time traffic routing	Designed for vehicular traffic only; not applicable to pedestrian tourist destination management; not localised for Sri Lanka

None of these platforms currently predict future crowd levels using flight arrivals or hotel occupancy. None provide real-time alerts when a tourist site is approaching overcrowding. None recommend eco-friendly alternative sites when popular destinations are heavily visited. These gaps form the focus of this research.

Proposed Solution

This research proposes a web-based Tourism Risk & Context Intelligence System that collects data from four sources, processes it through a Random Forest machine learning pipeline, and delivers four practical capabilities to tourists via a web interface: contextual risk classification (Low / Medium / High), real-time overcrowding alerts, safe time recommendations, and automatic itinerary adjustment with green site redirection. This component is one of four within the group's AI-Based Tourism Safety and Itinerary Platform (R26-IT-152), alongside the SOS Emergency Detection System (De Silva D.S.K), AI Itinerary Generator (Wanniarachchi P.M.R), and Hotel-Flight Driver Matcher (De Silva D.C.M). All components share a unified database and communicate through REST APIs.

SDG Alignment

This project aligns with three of the United Nations Sustainable Development Goals:

1. SDG 11 – Sustainable Cities and Communities (Target 11.4): The green site feature and crowd intelligence capability will direct tourist demand away from over-utilised heritage sites, reducing structural and environmental damage.
2. SDG 8 – Decent Work and Economic Growth: Improving the tourist experience through risk-aware advisory services will increase tourist spending and indirectly support sustainable employment in the tourism sector.
3. SDG 9 – Industry, Innovation and Infrastructure: The platform provides digital tourism infrastructure that currently does not exist in Sri Lanka and has not kept pace with regional competitors.

Research Cluster Alignment

This project is aligned with the AI Centre of Excellence (CEAI) at SLIIT and specifically under the research cluster of Machine Learning and Data Science for Real World Decision Making.

1.2 Literature Review

Ten scholarly sources were reviewed, covering crowd and visitor prediction at tourist sites, machine learning for demand forecasting, overtourism management, sustainable tourism, contextual risk assessment, and the Sri Lankan tourism context. Table 2 provides a structured summary of the reviewed sources.

Table 2: Summary of Literature Review

#	Reference	Method	Key Finding	Gap Identified
1	Bravo et al. (2023) [2]	Linear Regression, KNN, Decision Tree, Random Forest	Linear Regression achieved best performance for heritage site visitor prediction in Peru; all four algorithms tested; 80/20 split; public data viable	Peru-specific; no alerts; no itinerary adjustment; no green site feature; no flight data
2	Kim et al. (2025) [3]	RF, Gradient Boosting, LightGBM with Bayesian optimisation	RF, GB, and LightGBM all viable for trail visitor prediction; tollgate traffic volume and weekend flags most significant predictors	Nature trail context only; no site-level alerts; no integration with tourist itinerary
3	Mihalič (2020) [4]	Theoretical sustainability framework	Overtourism requires real-time demand-side management tools; supply-side policy alone is insufficient	Purely theoretical; no implemented prediction system; no technical solution
4	Santos-Rojo et al. (2023) [5]	Bibliometric analysis	Overtourism research has grown significantly since 2018; applied technical implementations underrepresented; European urban areas dominate research	No technical system implemented; no developing-country or island tourism context
5	Atrio Lema et al. (2025) [6]	SARIMA, TBATS time-series	Pilgrim flow can be predicted using seasonal time-series at pilgrimage route hotspots	Time-series only; no ML ensemble; no real-time alert; no flight or hotel data
6	Nurhasanah et al. (2022) [7]	SARIMA	SARIMA effectively models seasonal patterns in international tourist arrivals at a national level	Country-level only; no site-specific prediction; no crowd density output
7	Breiman (2001) [8]	Random Forest algorithm design	RF reduces variance through bagging over multiple trees; handles nonlinear relationships and high-dimensional tabular data effectively	Foundational algorithmic paper; no tourism application

#	Reference	Method	Key Finding	Gap Identified
8	SLTDA (2024) [1]	Official national statistics	2.36M arrivals; 1,600+ guide shortfall; USD 148 avg spend; 500+ registered sites with coordinates	Annual aggregate data only — no hourly or daily site-level granularity
9	Bollenbach et al. (2025) [9]	Random Forest, knowledge graphs	RF combined with route similarity features improves sustainable tourism routing	Route similarity focus only; no crowd density; no alert system
10	He et al. (2022) [10]	CNN-GRU neural networks	Deep learning achieves high accuracy for citywide crowd prediction but requires large fine-grained datasets	Large-dataset requirement; city-level not site-level; no tourist-specific features

Discussion of Existing Research Approaches

Bravo et al. [2] provide the most directly applicable methodological foundation for this work. Their study applies four supervised regression algorithms to predict visitor numbers at heritage tourist attractions in Peru, using public official data with an 80/20 train-test split in Python. Their validation confirms that standard supervised ML is technically feasible and academically appropriate for this class of prediction problem. However, the study operates in a Peru-specific context, uses no flight data, produces no real-time output, and does not classify risk levels.

Kim et al. [3] extend the methodological evidence base by demonstrating that Random Forest, Gradient Boosting, and LightGBM with Bayesian hyperparameter optimisation reliably predict daily visitor counts at natural trail sites using a multi-source feature set including weather, social media activity, and traffic volume. Their SHAP analysis — identifying traffic volume and day-of-week as the strongest predictors — informs the feature engineering strategy adopted here, particularly the use of flight arrival volumes and temporal features as primary predictors.

Mihalič [4] provides the theoretical foundation for the green site recommendation and context-aware risk features. Her sustainability framework argues that overtourism results from a demand-supply imbalance requiring active real-time management rather than purely supply-side policy, directly justifying the dynamic itinerary adjuster and risk-informed green site redirection as novel technical implementations of a well-supported theoretical intervention.

Santos-Rojo et al. [5] conducted a bibliometric analysis of 188 papers published between 2018 and 2021, finding that applied technological implementations remain rare and that most research has focused on European urban contexts. This creates significant geographic and contextual gaps directly addressed by this Sri Lanka-focused study.

Atrio Lema et al. [6] demonstrate the feasibility of site-specific visitor flow prediction using seasonal time-series models. Their work is instructive because it faces similar long-term aggregated data constraints and achieves usable prediction accuracy through seasonal modelling.

However, it is limited to time-series approaches and does not incorporate ML ensembles, contextual factors, or real-time risk alerts.

The Random Forest algorithm, developed by Breiman [8], combines bootstrapping and random feature selection to reduce prediction variance compared to a single decision tree. This property makes it well-suited to moderate-sized datasets such as the SLTDA records available for this research. He et al. [10] demonstrate that deep learning can exceed Random Forest accuracy for crowd prediction, but only when large fine-grained spatio-temporal datasets are available — which is not the case for Sri Lankan tourist sites at present. Bollenbach et al. [9] further confirm the contemporary relevance of Random Forest for sustainable tourism ML applications as recently as 2025.

Nurhasanah et al. [7] and the SLTDA annual report [1] together frame the data context. Nurhasanah et al. demonstrate SARIMA effectiveness at the national level, but without site-specific or risk-level outputs. The SLTDA report highlights the scale of the problem — 500+ registered sites, 2.36 million arrivals, 1,600+ guide shortfall — and the key data limitation: all official visitor data is aggregated annually with no hourly or daily site-level breakdown.

1.3 Research Gap

The review of existing literature highlights five key research gaps, each directly addressed by this component.

Gap 1 — Absence of flight arrival data as a predictor for site-level tourist risk and crowd forecasting: Every reviewed study using external demand signals relies on domestic road traffic sensors [3], internet search trends [2], or social media activity [9]. No study has used international flight arrival volumes as a site-level predictor. In Sri Lanka, the majority of international tourists enter through Bandaranaike International Airport (CMB), making hourly flight volumes a strong and measurable proxy for national tourist demand over the following 12–48 hours.

Gap 2 — Lack of context-aware risk intelligence systems for Sri Lankan tourist sites: The literature confirms that ML-based crowd prediction is technically feasible [2, 3] and that overtourism is a documented problem in Sri Lanka [1, 4]. Yet no system combining risk classification, contextual factors (events, holidays, weather), and crowd prediction has been developed or deployed in Sri Lanka. Existing platforms provide only static, historical information with no predictive or risk-scoring capability.

Gap 3 — No implemented green site demand redistribution system: Mihalič [4] and Santos-Rojo et al. [5] both argue that redirecting tourists to lower-footfall, eco-friendly alternatives is a practical response to overtourism. However, neither study — nor any other reviewed — implements this as a working technical system integrated with a live prediction pipeline.

Gap 4 — No dynamic itinerary adjustment based on real-time risk and crowd predictions: Kim et al. [3] and Atrio Lema et al. [6] generate visitor flow predictions but do not link these to tourists' planned itineraries. No existing tool automatically reorders a tourist's route in response to live or predicted risk and crowd conditions. This component introduces this capability for the first time in the Sri Lankan context.

Gap 5 — No system incorporates post-visit user feedback to improve prediction: None of the reviewed systems include post-visit feedback in a model retraining pipeline. This component addresses that through a continuous improvement approach where real trip data and user ratings progressively improve prediction accuracy — something generally absent from current studies.

Together, these five gaps position this research as a novel contribution at the intersection of machine learning, context-aware risk intelligence, and sustainable tourism management in Sri Lanka.

2 OBJECTIVES

2.1 Main Objectives

To design, develop, and evaluate a web-based Tourism Risk & Context Intelligence System that integrates flight arrival data, hotel occupancy records, historical visitor statistics, local event calendars, and weather conditions into a Random Forest machine learning pipeline in order to classify risk levels at Sri Lankan tourist sites, recommend optimal visiting times, generate overcrowding and risk alerts, dynamically adjust tourist itineraries, and promote eco-friendly alternative sites — thereby improving tourist safety and experience while supporting sustainable tourist flow management in Sri Lanka.

2.2 Specific Objectives

1. SO1 — Data Acquisition: Collect, integrate, and store multi-source datasets from the AviationStack API (international flight arrivals at CMB), the group's hotel dataset (approximately 10,000 records), SLTDA visitor statistics, public holiday and local events data, and OpenWeatherMap API (rainfall and temperature), covering approximately 500 registered Sri Lankan tourist sites.
2. SO2 — Model Development: Build a Random Forest model using features including airport flight arrival volume, regional hotel occupancy indicators, temporal indicators (day of week, public holidays, festival periods), and weather conditions to predict site-level crowd density and classify risk level (Low / Medium / High). Model accuracy will be measured using MAE and R^2 against a simple historical average baseline.
3. SO3 — Risk Classification and Safe Time Recommendation: Implement a risk scoring engine on the web interface that classifies each tourist site into Low, Medium, or High risk categories for a given time window and recommends time windows with lower predicted crowd and risk conditions based on user-specified visit dates.
4. SO4 — Overcrowding and Risk Alert System: Develop a crowd and risk monitoring engine that generates real-time alert notifications via Server-Sent Events (SSE) when a site in the tourist's planned itinerary is predicted to reach high crowd or high risk levels within the visit window.
5. SO5 — Dynamic Itinerary Adjustment and Green Site Redirection: Design an adaptive itinerary adjustment module that reorganises planned stops based on predicted crowd and risk levels, and recommends environmentally responsible alternative sites when congestion or risk is forecast to exceed acceptable thresholds.
6. SO6 — Feedback and Retraining Pipeline: Implement an anonymous post-visit rating system through the web interface. Collected data will be used to retrain the prediction model periodically, progressively improving risk and crowd prediction accuracy over time.

3 METHODOLOGY

Research Approach

This research follows a Design Science Research (DSR) methodology, which is a well-established framework for research that aims to design and evaluate a technical artefact to solve a real-world problem [2, 3]. The DSR cycle proceeds through five phases: problem identification and motivation, objective definition, artefact design, artefact development, and evaluation. The artefact in this research is the Tourism Risk & Context Intelligence System web module.

Each phase produces deliverables that are evaluated against defined criteria before the next phase begins, enabling iterative refinement throughout the development timeline.

Technical Methods and Algorithms

The primary predictive technique is supervised machine learning via Random Forest regression and classification. Algorithm selection was made through a structured evaluation of four candidate approaches.

Linear Regression was considered and discarded because it presupposes a linear relationship between input features and crowd or risk output. The relationships among flight volume, hotel occupancy, festival periods, weather conditions, and visitor patterns are well understood to be nonlinear — particularly during peak seasons and holiday periods — making a linear model an inappropriate fit for this problem.

Neural Networks and deep learning approaches were evaluated and not selected. He et al. [10] demonstrate that deep learning models such as CNN-GRU can achieve high accuracy for crowd prediction, but they require large, fine-grained spatio-temporal datasets that are not currently available for Sri Lankan tourist sites. Applying deep learning to a small or simulated dataset would lead to overfitting and unreliable predictions.

Support Vector Regression was reviewed but not selected due to increased computational complexity, lower interpretability compared to Random Forest, and high sensitivity to feature scaling across multiple heterogeneous data sources [9].

Random Forest was selected as the most appropriate model for the following reasons: it handles nonlinear relationships in mixed-type feature data effectively; it performs well on moderate-sized datasets; it provides SHAP-interpretable feature importance scores that help explain which inputs most influence risk and crowd predictions; and it has been validated for tourist visitor prediction in comparable contexts by Bravo et al. [2] and Kim et al. [3], and for sustainable tourism routing by Bollenbach et al. [9] as recently as 2025.

Model Training and Validation

The dataset will be divided into training and testing sets using an 80/20 split, following the protocol used by Bravo et al. [2]. A baseline model will first be constructed using historical averages for day-of-week and month. Model performance will then be assessed against this baseline using Mean Absolute Error (MAE) and R^2 score on the held-out test set, with the Random Forest model expected to outperform the baseline on both metrics. Hyperparameter tuning will be conducted via grid search using 5-fold cross-validation, evaluating parameters including number of trees and maximum tree depth.

Evaluation metrics will include: prediction accuracy of risk classifications; alert generation accuracy against held-out test cases; and field-based observational validation at selected tourist sites, comparing predicted risk and crowd levels against directly observed conditions.

Tools and Platforms

Table 3: Technology Selection and Justification

Component	Selected Technology	Justification	Alternatives Rejected
ML Development	Python, scikit-learn, pandas, NumPy	Industry standard tools for machine learning and data analysis. Random Forest regression is natively implemented in scikit-learn, and pandas/NumPy support efficient preprocessing of tabular datasets.	R (less convenient integration with web APIs); MATLAB (proprietary and higher licensing cost)
Backend Framework	Flask (Python)	Lightweight Python web framework suitable for building REST APIs and integrating directly with the scikit-learn prediction pipeline.	Django (higher overhead for an API-focused backend); Node.js (separate ML environment required)
Frontend Framework	React.js	Component-based architecture supports interactive dashboards, dynamic data updates, and responsive display of alerts and crowd predictions.	Plain HTML/JS (less maintainable for complex interactive interfaces); Angular (steeper learning curve and heavier framework overhead)
Flight Data	AviationStack API	Provides real-time and historical flight schedule data for arrivals at Bandaranaike International	FlightAware (higher cost); manual web scraping

Component	Selected Technology	Justification	Alternatives Rejected
		Airport (CMB), which can be used as a proxy indicator for international tourist inflow.	(unreliable and legally uncertain)
Weather Data	OpenWeatherMap API	Free tier provides historical weather observations and short-term forecasts (temperature and rainfall) suitable for feature engineering in crowd prediction.	Dark Sky (discontinued); Weatherstack (more limited free-tier historical coverage)
Hotel Data	Booking.com dataset + Group DB	The group already maintains a dataset containing approximately 10,000 hotel records. Regional aggregation of occupancy indicators can be derived from this dataset for demand estimation.	Airbnb data (restricted access); Agoda API (limited historical coverage)
Historical Visitor Data	SLTDA annual reports	Official national tourism statistics source providing historical visitor trends and registered tourist site information.	Web scraping TripAdvisor (unreliable, legally uncertain)
Real-time Alerts	WebSocket (server-sent events)	Enables lightweight real-time server-to-browser notifications for crowd alerts without requiring full page reloads.	Polling (higher server load); Firebase (third-party dependency and additional cost)
Integration	REST API	Enables structured communication between system modules using predefined endpoints and JSON responses.	GraphQL (unnecessary complexity for defined data contracts)
Database	PostgreSQL	Relational database suitable for structured tabular data and capable of supporting spatial queries for location-based tourist site filtering.	MongoDB (less suitable for structured relational datasets); SQLite (limited multi-user concurrency)

Data Collection and Ethical Considerations

Data used in this study was accessed either through publicly available sources, official publications, or datasets previously compiled by the research group. All external data sources (AviationStack API and OpenWeatherMap API) were accessed per their developer services and usage policies. No personally identifiable information (PII) from tourists is collected or processed at any stage of the research.

Post-visit feedback collected through the web interface is anonymised at the point of submission before being stored in the system database. Feedback records contain only general satisfaction ratings and optional comments, and do not include identity details, device identifiers, or location tracking information.

Hotel-related data is analysed only at an aggregated regional level. Individual booking records or customer-level transaction data are not accessed or stored. The green site recommendation catalogue is constructed using publicly available tourist site information and classifications

published by SLTDA. No scraping or collection of personal content from social media platforms is conducted as part of this research.

All data assumptions and limitations are fully documented throughout the modelling phase. Refer to Appendix E for further information on ethical issues related to data handling.

4 High level Architecture

Overview

The Tourism Risk & Context Intelligence System follows a three-layer web application architecture that separates data ingestion and preprocessing, risk prediction and intelligence processing, and user-facing web output. The system is accessed by tourists and tourism operators through a standard web browser.

Risk prediction and crowd scoring are performed on the server side. Prediction results are generated periodically and stored in a cache for efficient retrieval. Results are delivered to the client through REST API endpoints, while real-time alert notifications are pushed using Server-Sent Events (SSE).

Component Interactions

Within the overall group platform, this module acts as the data analytics, risk prediction, and context intelligence component. The risk score, crowd level, and alert flag API endpoints produced by this system are consumed by:

— The AI Itinerary Generator (Wanniarachchi P.M.R) to produce crowd-aware and risk-aware travel plans. — The SOS Emergency Detection System (De Silva D.S.K) to enhance situational awareness by incorporating crowd density and risk indicators. — The Hotel-Flight Driver Matcher (De Silva D.C.M) to avoid congested or high-risk pickup and drop-off zones.

Completed itinerary plans generated by the itinerary module can be passed back to this system so that the Dynamic Itinerary Adjuster can re-evaluate and reorder planned stops when predicted risk or crowd levels change. All modules communicate through predefined REST API endpoints, allowing each component to remain independently testable while sharing data across the platform.

Data Flow

Data enters the system through internal and external sources via scheduled ingestion tasks. Sources include flight arrival data, weather observations, hotel occupancy indicators, historical tourism statistics, and public holiday/event calendars.

Raw data is processed through a data cleaning and feature engineering pipeline, producing relevant attributes for the model: time of day, day of week, weather conditions, local event flags, and tourism demand indicators.

Feature vectors pass through the Random Forest prediction model to estimate crowd levels and classify risk at registered tourist sites for specified time intervals. The risk score and crowd level are cached and routed to:

— The Risk Classification Engine, which categorises output as Low, Medium, or High — The Recommendation Engine, which identifies optimal visiting time windows — The Alert Engine, which notifies users when predicted conditions exceed safe thresholds — The Dynamic Itinerary Adjuster, which reorders planned visits based on live risk predictions

Post-visit feedback submitted through the web interface is anonymised and stored for periodic model retraining.

Scalability and Security Considerations

Crowd and risk predictions are generated on a batch schedule so that results can be reused across multiple concurrent user sessions, reducing server load during peak tourism periods rather than recalculating per individual request.

All external API credentials are securely stored on the server using environment variables and are never exposed to client-side code. User feedback is stored in anonymous format (without personal identifiers or session information) to comply with data privacy principles. The layered REST-based architecture ensures that failures in individual modules do not propagate across the platform.

5 User Requirements

Stakeholder Identification

Three main stakeholder groups are identified.

International tourists represent the primary end users, interacting with the web interface to access predicted risk and crowd conditions, receive alerts, and follow dynamically adjusted travel plans.

Tourism operators and hotels are secondary users who may use the platform dashboard to observe general crowd trends, risk conditions, and demand patterns around nearby tourist sites.

Government organisations, including SLTDA, are tertiary stakeholders who may benefit from aggregated analytics for tourism flow monitoring and long-term policy planning

Table 4: Functional Requirements

FR#	Requirement Description	Priority	Stakeholder
FR1	The system shall ingest and store data from the AviationStack API, the group hotel dataset, SLTDA visitor records, and OpenWeatherMap API for registered Sri Lankan tourist sites used in the project dataset.	High	System
FR2	The system shall generate crowd level predictions using a continuous crowd score that is later categorised as Low, Medium, or High for display purposes.	High	Tourist
FR3	The web interface shall display recommended visiting time windows for each tourist site within a 24-hour horizon, ranked by predicted crowd score.	High	Tourist
FR4	The system shall generate an alert notification for the user's active web session when a site in the user's itinerary is predicted to reach high crowd levels within a defined time window.	High	Tourist
FR5	The Dynamic Itinerary Adjuster will indicate a revised visit order and recommend an alternate place to visit based on observed crowd levels during time of visit.	High	Tourist
FR6	The online portal will feature a Green Sites section that highlights alternative destinations proven to have lower visitor footfalls.	Medium	Tourist
FR7	The system shall collect anonymous post-visit feedback ratings from users through the web interface and store responses in the database for use in model retraining.	Medium	Tourist

FR#	Requirement Description	Priority	Stakeholder
FR8	The system will provide the predicted crowding for facilities as well as alert notifications to be used by additional modules in the overall site group by means of defined RESTful API endpoints.	High	System
FR9	The platform shall provide a dashboard view showing general crowd trends for selected tourist sites.	Medium	Operator
FR10	The system shall store anonymised feedback data and support periodic model retraining using accumulated data.	Medium	System

Table 5: Non-Functional Requirements

NFR#	Category	Measurable Requirement
NFR1	Performance	Crowd prediction results shall render on the web interface within 5 seconds of a user request under typical load conditions.
NFR2	Accuracy	The prediction model should aim to outperform a simple historical-average baseline using MAE and R^2 evaluation metrics.
NFR3	Accuracy	The prediction ability (both precision and recall) for crowd an overcrowding alert will be maintained, at a minimum of > or = 0.80, on test cases held-out of the training set.
NFR4	Scalability	The systems must have the ability to generate predictions for multiplicity of tourist attractions, without introducing unacceptable degradation in the predicted performance, during the maximum usage time period.
NFR5	Availability	The web application must remain operable through the peak period of visitation by tourists, provided that the usual maintenance/testing timelines are met.
NFR6	Usability	The web interface should function correctly on both desktop and mobile browsers.
NFR7	Privacy	No personally identifiable tourist data shall be collected; all feedback records shall be stored in anonymised form.
NFR8	Maintainability	The prediction model shall support periodic retraining and version updates without requiring major changes to the rest of the system.

6 Commercialization plans

Target Market and Customer Persona

The potential user base includes three main segments.

The primary B2C segment consists of international tourists visiting Sri Lanka. According to SLTDA statistics, approximately 2.36 million international tourists visited Sri Lanka in 2025, with national targets aiming to reach around 3 million arrivals by 2026 [1]. Independent travellers and family groups are the most likely users of a risk and crowd intelligence platform, as they typically plan their own routes and are most sensitive to safety and congestion at popular attractions.

The secondary B2B segment includes tourism service providers such as hotels, guesthouses, tour operators, and local activity providers registered with SLTDA. Access to risk and crowd intelligence can help this group make staffing plans, determine touring schedules, and adapt visitor services proactively.

A third stakeholder group consists of government and tourism management bodies such as SLTDA, which can use aggregated risk and visitation data to support tourism flow planning and policy decisions.

Value Proposition

For tourists, the system provides crowd-level and risk classifications for specific visiting times, enabling them to plan safer, more enjoyable itineraries rather than arriving at over-crowded or high-risk sites without warning.

For tourism operators, the system produces general intelligence about visitor movement and demand patterns that can inform staffing decisions, guided tour scheduling, and promotion of alternative destinations during peak congestion periods.

Revenue Model

Phase 1 – Basic Freemium Model: General risk and crowd forecasts and visit recommendations are provided free of charge. Advanced features such as real-time alert notifications, detailed risk breakdowns, and advanced itinerary adjustment may be offered for a small per-trip or subscription fee.

Phase 2 – Tourism Operator Dashboard: A subscription-based dashboard could be introduced for tourism operators requiring aggregated risk trend information for selected tourist areas, with pricing varying by feature access and usage level.

Phase 3 – Tourism Analytics Insights: If the platform accumulates significant aggregated data, anonymised risk and crowd insights could potentially be licensed to tourism planning organisations or research institutions.

Competitive Advantage

The proposed system offers several advantages compared with currently available tourism information tools. First, it focuses specifically on predicting risk and crowd conditions at tourist sites — a capability absent from all existing platforms used in Sri Lanka. Second, it synthesises multiple data types — flight arrivals, weather, events, and tourism statistics — into a unified intelligence layer. Third, it is integrated into the broader group platform, which includes itinerary planning, safety monitoring, and transport coordination modules, enabling more complete travel planning assistance.

Projected Revenue (Conceptual Estimate)

As a conservative estimate, if the platform achieves 5,000 monthly active tourist users with a 10% premium conversion rate at LKR 500 per trip, this would generate approximately LKR 250,000 per month in B2C revenue. An additional 20 tourism operator subscriptions at LKR 10,000 per month would contribute LKR 200,000, bringing projected monthly revenue to approximately LKR 450,000 — sufficient to cover ongoing API and hosting costs within the first operational year.

Intellectual Property Considerations

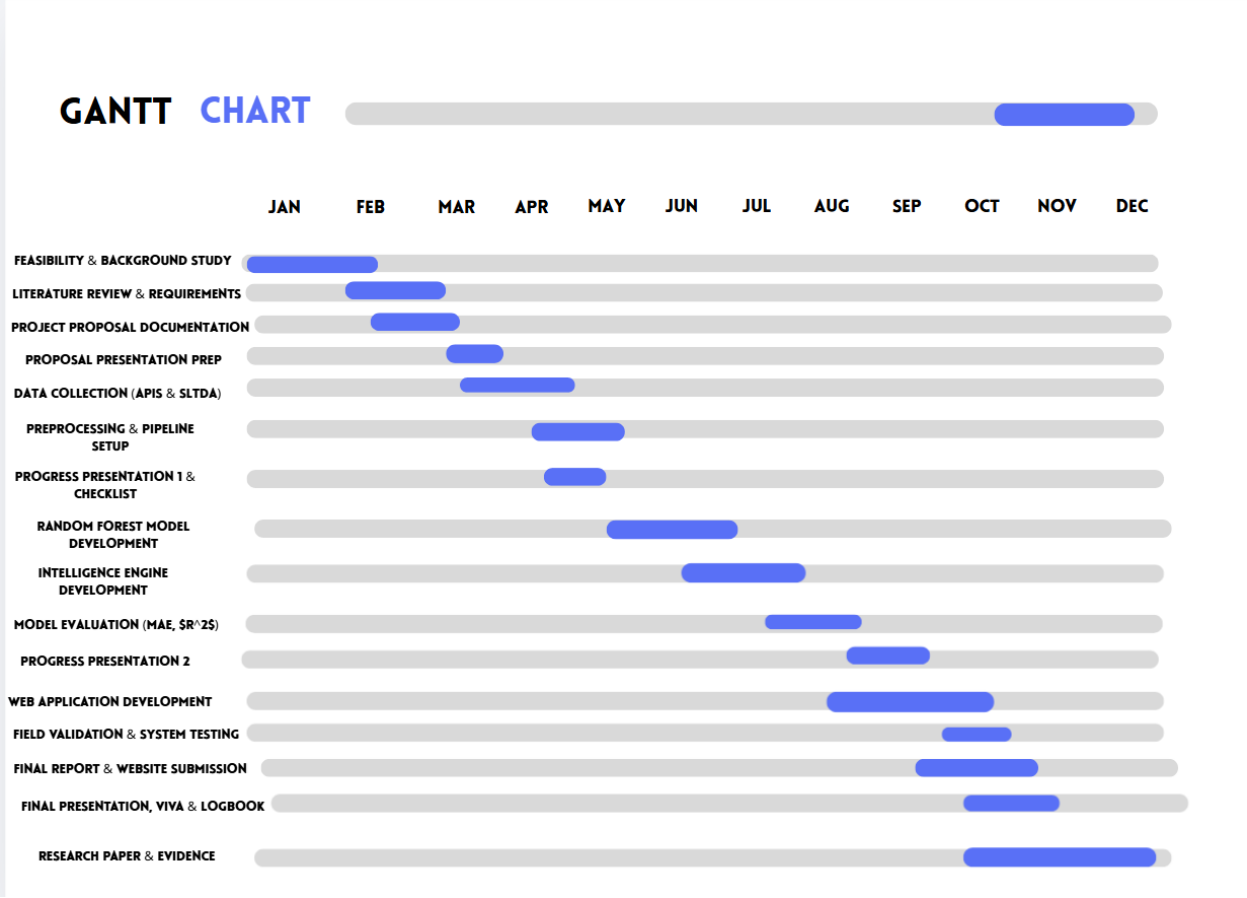
Several intellectual assets may emerge from this project.

The trained Random Forest prediction model developed using tourism-related datasets represents a research artefact that demonstrates the feasibility of crowd forecasting for tourist destinations.

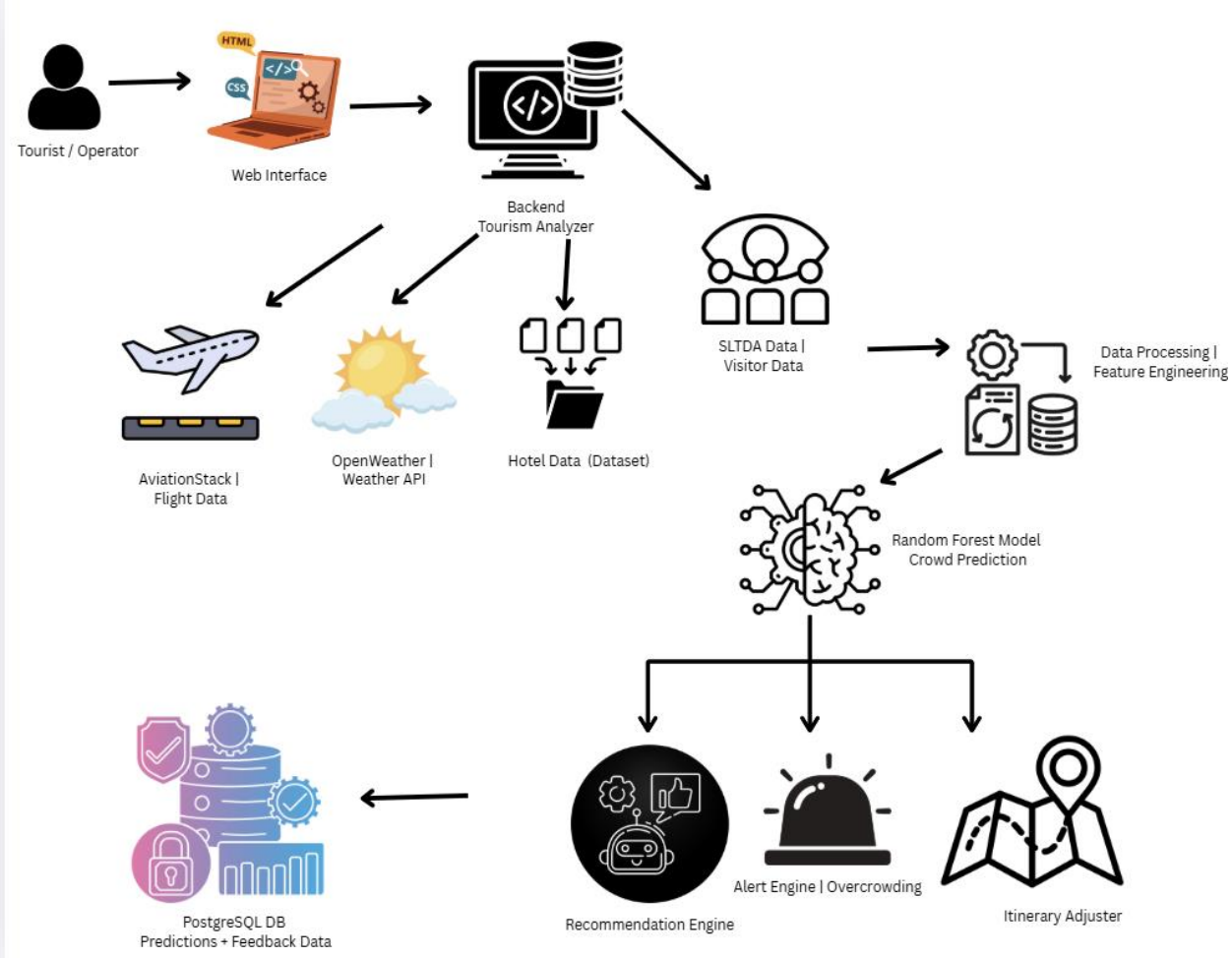
The tourist site dataset and green site catalogue, compiled using publicly available information from the Sri Lanka Tourism Development Authority and other sources, may also form a structured dataset useful for tourism analytics research.

In addition, anonymised post-visit feedback data collected through the platform could become a valuable dataset for improving future prediction models and analysing tourist experience trends.

7 Gantt chart



8 System diagram



9 Conclusion

This proposal presents the Tourism Risk & Context Intelligence System, a web-based risk prediction and contextual advisory platform designed to address a clear and documented gap in Sri Lanka's tourism infrastructure. By integrating flight arrival data from AviationStack, regional hotel occupancy indicators, historical SLTDA visitor statistics, public holiday and event data, and OpenWeatherMap weather data into a Random Forest machine learning pipeline, the system delivers four practical capabilities — risk-level classification, real-time overcrowding alerts, safe time recommendations, and eco-friendly site redirection — across 500+ registered tourist destinations.

The research addresses five specific gaps identified in the reviewed literature: the absence of flight arrival data as a site-level risk and crowd predictor; the lack of any ML-based context-aware risk intelligence system deployed in Sri Lanka; the non-existence of an implemented green site demand redistribution tool; the absence of real-time itinerary adjustment based on live risk and crowd predictions; and the omission of post-visit feedback loops in existing prediction systems.

The proposed methodology, grounded in Design Science Research and validated against comparable tourism ML studies, is technically feasible within the defined scope and timeline. If the target accuracy benchmarks of MAE ≤ 0.25 and $R^2 \geq 0.7$ are achieved, the system will represent a novel and deployable contribution to sustainable tourism management in Sri Lanka, with direct alignment to SDG 8, SDG 9, and SDG 11, and a credible pathway toward commercialisation through freemium, B2B SaaS, and data licensing revenue models.

10 BUDGET AND BUDGET JUSTIFICATION

Table 6: Budget Breakdown and Justification

Item	Description	Cost (LKR)	Justification
API Subscriptions	AviationStack Starter Plan (~USD 29/month × 3 months) for historical CMB flight data; OpenWeatherMap free tier for weather	9,600	AviationStack paid plan required for historical data access beyond 3 months and for batch query volumes needed for 500+ site matching; OpenWeatherMap free tier fully sufficient for development scope
Cloud Hosting	Backend server instance for Flask prediction service and React web application (3-month development and testing period)	11,500	Minimum viable cloud compute required to run Python ML prediction service and serve the web application; local hosting insufficient for group integration testing
Development Tools	Python, scikit-learn, pandas, NumPy, React (all open source); Jupyter Notebook for ML experimentation	3,000	Covers environment setup and tooling; all primary ML and frontend libraries are open source at zero licensing cost; cost covers any proprietary IDE licences and deployment tooling
Field Validation	Transport and accommodation for site visits to 3–5 tourist locations for ground-truth crowd condition recording	15,000	Essential for real-world model validation; only method available to obtain ground-truth crowd conditions for specific sites and specific times to compare against model predictions
Miscellaneous	Documentation printing, report binding, contingency for unexpected API costs	4,000	Standard buffer for academic project; covers final submission printing and any AviationStack overage costs
Total		43,100	

Note: AviationStack pricing verified at aviationstack.com as of January 2026. All ML and frontend libraries are open source. Group members share API credentials to avoid duplication of subscription costs. Hosting costs are shared proportionally with the group where infrastructure is shared.

11 WORK BREAKDOWN STRUCTURE

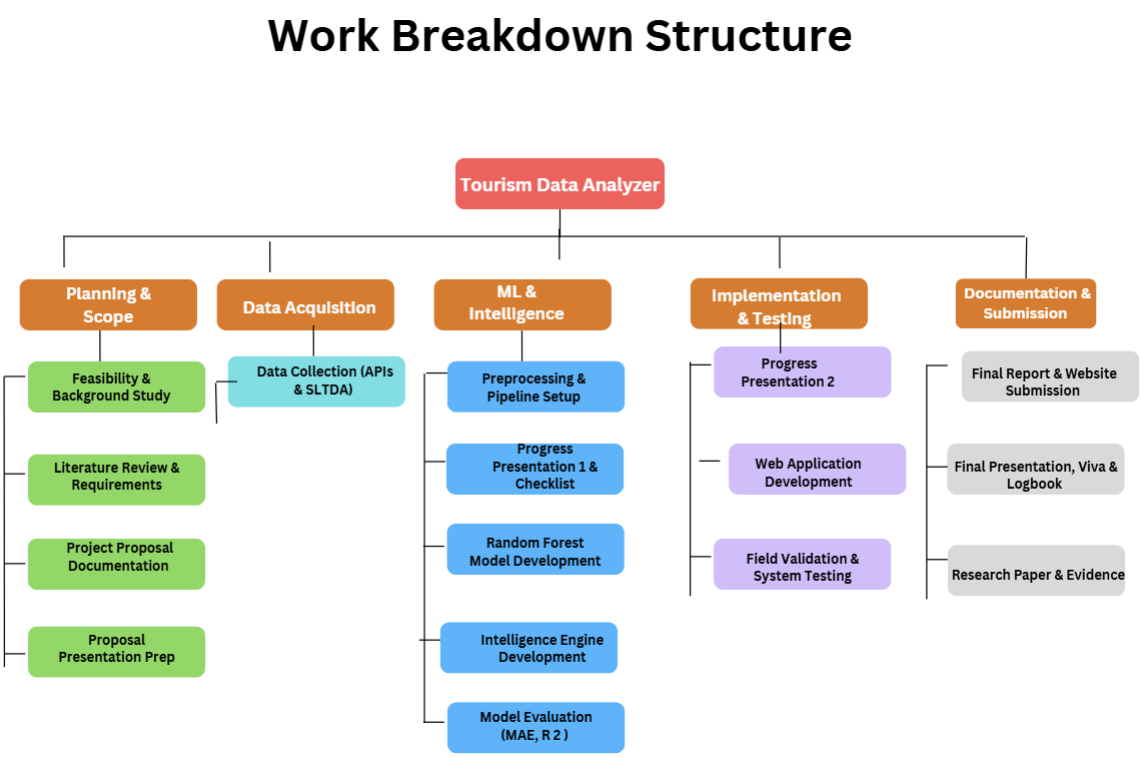


Table 7: Work Breakdown Structure

Task ID	Task	What Gets Delivered	When
T1.1	Connect AviationStack API and download available CMB arrival flight data	Flight arrival dataset with timestamps and origin airports	March 2026
T1.2	Connect OpenWeatherMap API and collect weather data for selected tourist site locations	Weather dataset per site per day	March 2026
T1.3	Review SLTDA reports and compile a database of registered tourist sites and available visitor indicators	Tourist site master dataset	March–April 2026

Task ID	Task	What Gets Delivered	When
T1.4	Extract hotel information from the group dataset and organise data by geographic region	Regional hotel demand indicators dataset	April 2026
T1.5	Build the green site catalogue using publicly available tourism classifications	Tagged list of alternative tourist destinations	April 2026
T2.1	Develop the preprocessing pipeline to clean, normalise, and align datasets	Clean dataset prepared for machine learning	April–May 2026
T2.2	Train the Random Forest model and perform hyperparameter tuning using grid search and cross-validation	Trained model with validation results	May 2026
T2.3	Evaluate model performance against a baseline using metrics such as MAE and R^2	Model evaluation report	May–June 2026
T3.1	Develop the Recommendation Engine to identify favourable visiting time windows	Functional recommendation feature	June 2026
T3.2	Implement the Overcrowding Alert Engine using server-sent event notifications	Working alert with test results	June 2026
T3.3	Develop the Itinerary Adjuster to reorder planned visits and suggest alternative sites when needed	Working adjuster module	June–July 2026
T3.4	Implement feedback collection, anonymisation, and periodic model retraining script	Feedback and retraining pipeline	July 2026
T4.1	Develop the React web interface including Site Explorer, Itinerary Page, and Green Sites Page	Full deployed website	August–September 2026
T4.2	Develop and document REST API endpoints for integration with other group modules	API documentation and endpoints	August 2026

Task ID	Task	What Gets Delivered	When
T4.3	Conduct limited field observations at selected tourist sites and compare observed crowd levels with predictions.	Field validation report	September 2026
T4.4	Prepare the final report and research paper for submission	Completed research documentation	October 2026

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13 APPENDICES

Appendix A — AviationStack API Sample Response Structure

The AviationStack API returns JSON objects for each arriving flight, with flight date, airline, scheduled arrival time, estimated arrival time, delay information, and origin airport. These records can be grouped to estimate daily arriving flight activity at Bandaranaike International Airport and used as a proxy indicator for tourist inflow patterns relevant to risk and crowd prediction.

Appendix B — SLTDA Site Data Sample

SLTDA's 2024 annual report lists 500+ registered sites with: name, province, district, GPS coordinates, category (UNESCO Heritage, Archaeological, Religious, Nature Reserve, Beach, Cultural), and total annual visitor statistics. These form the site master database and primary historical visitor feature. Where hourly data is unavailable, simulated distributions are used during development and clearly labelled as simulated in all evaluation reports.

Appendix C — Preliminary Web Interface Mockups

Three tourist-facing pages are planned:

Site Explorer: Search bar with site name and province filter. Each site card includes a colour-coded risk badge (Green = Low, Amber = Medium, Red = High), a Best Times badge showing the top three visiting windows, and a Save to Itinerary button.

Itinerary Page: Shows a list of planned sites with visit times. Each entry displays the predicted risk badge for the planned time. When an alert fires, a banner appears: "Sigiriya is forecast to be High Risk at your planned time — visit at 7am or switch to [Green Site]." One-click Reorder and Switch buttons are available.

Green Sites Page: Gallery of SLTDA eco-tagged sites with lower risk and crowd scores. Sites suggested by the Itinerary Adjuster appear with a Recommended Alternative label.

Appendix D — Risk Analysis Table

Risk	Likelihood	Impact	Mitigation
SLTDA data is annual totals only	High	Medium	Use simulated hourly data for 20–30 key sites; validate through field visits; document clearly
AviationStack API exceeds budget	Low	Medium	Cache all historical data after first download; batch queries within plan limits

Risk	Likelihood	Impact	Mitigation
Prediction accuracy lower than expected	Medium	High	Expand feature set including events/holidays; reduce initial scope to 20–30 sites; document limitation honestly
Group REST API integration delayed	Low	Medium	Agree API schema in March 2026; each component independently testable with mock data
Field validation logistics issues	Low	Low	Plan visits during shoulder season; three sites minimum; budget set aside
Tourist adoption below break-even	Medium	Low	Conservative 5,000 MAU assumption; B2B revenue provides second path to break-even

Appendix E — Ethical Clearance Considerations

No personally identifiable tourist data is collected or stored at any stage. Post-visit feedback is voluntary, anonymised at submission before storage, and contains no identifiers or location tracking. Hotel occupancy data is used only as regional aggregate averages — no individual bookings are accessed. The green site catalogue uses only publicly available SLTDA classifications. The risk prediction model is audited periodically to detect bias against particular site types, regions, or visitor groups. All practices comply with Sri Lankan data protection guidelines and SLIIT undergraduate research ethics policies.

Appendix F — Survey Instruments for Field Validation

A structured observation checklist will be used during site visits scheduled for September 2026 at three to five tourist sites. The instrument will collect: site name, date and time of observation, crowd level on a five-point scale (1 – very empty to 5 – extremely crowded), estimated risk level (Low / Medium / High based on observed conditions), weather conditions, and presence of queues or congestion indicators.